Calculus Practice — Derivatives through the Chain Rule

8 questions. Try all of them before checking the answers below.

Practice Questions

Q1 Find the derivative of:

$$f(x) = e^{3x^2+1}$$

Q2 Find the derivative of:

$$f(x) = 3x^4 + \frac{2}{x^2} - 5\sqrt{x}$$

Q3 The sigmoid function is defined as:

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

Show that:

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$

Q4 Find the derivative of:

$$f(x) = \frac{2x+1}{x^2+3}$$

Q5 Find the derivative of:

$$f(x) = 4x^5 - 3x^2 + 8$$

Q6 Find the derivative of:

$$f(x) = \ln(5x^2 - 3x)$$

Q7 Find the derivative of:

$$f(x) = (x^2+1)(3x-2)$$

Q8 Find the derivative of:

$$f(x) = (4x^3 - 2x)^5$$

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Answers & Solutions

A1. Let $u = 3x^2 + 1$, so $f(x) = e^u$ and $u' = 6x$.

$$f'(x) = 6x\, e^{3x^2+1}$$

A2. Rewrite as $f(x) = 3x^4 + 2x^{-2} - 5x^{1/2}$, then differentiate term by term.

$$f'(x) = 12x^3 - \frac{4}{x^3} - \frac{5}{2\sqrt{x}}$$

A3. Rewrite $\sigma(x) = (1 + e^{-x})^{-1}$. Let $u = 1 + e^{-x}$, so $\sigma = u^{-1}$ and $u' = -e^{-x}$.

$$\sigma'(x) = -u^{-2}u' = \frac{e^{-x}}{(1 + e^{-x})^2}$$

Since $1 - \sigma(x) = \dfrac{e^{-x}}{1 + e^{-x}}$, this is exactly $\sigma(x)(1 - \sigma(x))$.

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$

A4. Quotient rule with $u = 2x + 1,\ v = x^2 + 3$:

$$f'(x) = \frac{2(x^2+3) - (2x+1)(2x)}{(x^2+3)^2} = \frac{-2x^2 - 2x + 6}{(x^2+3)^2}$$

$$f'(x) = \frac{-2(x^2 + x - 3)}{(x^2+3)^2}$$

A5.

$$f'(x) = 20x^4 - 6x$$

A6. Let $u = 5x^2 - 3x$, so $f(x) = \ln(u)$ and $u' = 10x - 3$.

$$f'(x) = \frac{10x - 3}{5x^2 - 3x}$$

A7. Product rule with $u = x^2 + 1,\ v = 3x - 2$:

$$f'(x) = (2x)(3x - 2) + (x^2 + 1)(3) = 9x^2 - 4x + 3$$

A8. Let $u = 4x^3 - 2x$, so $f(x) = u^5$ and $u' = 12x^2 - 2$.

$$f'(x) = 5(4x^3 - 2x)^4(12x^2 - 2) = (60x^2 - 10)(4x^3 - 2x)^4$$